

Person Detection, Tracking, and Action Recognition on TITAN

Zouhair Adam Hamaimou

University of Toronto Institute for Aerospace Studies

zouhair.hamaimou@mail.utoronto.ca

Abstract—We study pedestrian detection, tracking, and multi-head action recognition on the TITAN urban-driving dataset. Our pipeline couples a single-class YOLO11 detector with ByteTrack for person tracking and a 3D ResNet-18 that predicts five TITAN action families from short person tubes. Under compute constraints, YOLO11 is fine-tuned on the full TITAN person split for one epoch and on a 10% subset for 30 epochs, and the action recognizer is trained and evaluated on tubes from a 20% clip subset built from ground-truth tracks. YOLO11 achieves strong person localization (box mAP₅₀ up to 0.987), showing that high-quality boxes are attainable. The R3D-18 recognizer reaches 0.83 macro accuracy but only 0.14 macro mAP over non-*none* classes, despite rare-class oversampling and class-balanced losses; extreme label imbalance across heads remains the main challenge.

Keywords: TITAN dataset, pedestrian detection, action recognition, multi-head classification.

I. INTRODUCTION

Urban-driving perception systems must reason not only about *where* pedestrians are, but also *what* they are doing over time. Actions such as biking, entering a building, talking on a phone, or pushing an object provide cues about short-term intent and therefore about future motion. The TITAN framework of Malla *et al.* [1] exploits this idea by feeding discrete pedestrian actions into an interaction encoder and motion decoders to improve trajectory forecasting. In such pipelines, the quality of the action-recognition module directly limits downstream performance.

Replicating or extending TITAN requires an action recognizer that (i) operates on person-centric video tubes, (ii) outputs one label for each of several mutually exclusive action families, and (iii) copes with highly imbalanced label distributions. The original work provides only a high-level description of this component. This report therefore focuses on the perception pieces needed for TITAN: single-class pedestrian detection and tracking, and a standalone multi-head pedestrian action recognizer.

We instantiate this stack with a YOLO11 detector [2] and ByteTrack tracker [3] for person localization, followed by a multi-head 3D ResNet-18 (R3D-18) backbone [4] that predicts the five TITAN action families (atomic, simple-context, complex-context, communicative, and transportive) from person tubes. Under realistic compute limits, YOLO11 is trained on the full TITAN person split for a single epoch and a 10% subset for 30 epochs, while action recognition is learned on tubes from a 20% clip subset built from ground-truth tracks, so that recognition performance can be evaluated independently.

Contributions. In this work we:

- Quantify head-wise label imbalance in the TITAN action families and construct tube-based splits for multi-head action recognition.
- Benchmark YOLO11 variants with and without fine-tuning for single-class pedestrian detection on TITAN under compute constraints.
- Train and evaluate a multi-head R3D-18 action recognizer on TITAN tubes, and assess the effect of rare-class-aware sampling, inverse-frequency class weighting, and focal loss on per-head and per-class performance.

A. Motivation and Related Work

Early video classification systems applied 2D convolutional networks frame by frame and aggregated predictions over time [5]. Two-stream architectures [6] introduced separate RGB and optical-flow branches, motivating joint modelling of appearance and motion. Subsequent work replaced 2D convolutions with 3D convolutions, as in C3D [7] and long-term temporal convolutions [8], enabling networks to learn spatio-temporal features directly from video volumes. I3D [9] inflates 2D architectures into 3D to leverage ImageNet pre-training, and 3D ResNets [4] add residual connections to scale depth and capacity for video action recognition.

Pedestrian detection and multi-object tracking are complementary components in driving perception. Single-stage detectors from the YOLO family have become standard choices for real-time person localization, while online trackers associate per-frame detections into trajectories in crowded scenes. ByteTrack [3] is particularly relevant here: it improves recall by retaining low-score detections during data association without requiring additional appearance features, making it suitable for real-time tracking pipelines.

TITAN [1] builds on these advances by annotating urban-driving videos with person-centric tracks and five action families and by using actions as priors for interaction encoding and motion forecasting. However, the original work only sketches the action-recognition module and does not release a reference implementation or a detailed analysis of its long-tailed, multi-head label distributions. Our study targets this missing perception component, applying modern 3D CNNs and imbalance-aware objectives such as focal loss [10] to characterize what action-recognition performance can be achieved on TITAN.

II. PROBLEM STATEMENT

We consider a forward-facing RGB camera mounted on a vehicle driving in urban environments. The camera produces a video stream $\{I_t\}$ sampled at 10Hz. The perception problem has two coupled components:

- 1) **Person detection and tracking:** For each frame, localize all visible pedestrians with bounding boxes and associate them over time into person-centric tracks.
- 2) **Tube-level multi-head action recognition:** For each short spatio-temporal tube corresponding to a single person, predict one action label for each TITAN action family.

Let a person track be represented by a sequence of bounding boxes over time. From each track we extract fixed-length windows of consecutive frames and crop around the person to form tubes; each tube thus encodes a short segment of motion for a single individual. The TITAN schema defines five action families (atomic, simple-context, complex-context, communicative, and transportive), each with a discrete label set that includes a special *none* class when no action of that type is active.

Given a dataset of tubes $\{\mathbf{x}_i\}$ and their corresponding label vectors

$$\mathbf{y}_i = (y_i^{(h)})_h, \quad y_i^{(h)} \in \mathcal{Y}^{(h)},$$

the goal is to learn a model that, for each tube $\mathbf{x}_i$, predicts one label per action family. The action-recognition task is therefore a multi-head, multi-class classification problem on person-centric video tubes, complicated by the fact that many action classes are extremely rare compared to common behaviours such as walking or standing. In our experiments, we evaluate this model on TITAN using tubes constructed from ground-truth person tracks so that recognition performance can be assessed independently of detection and tracking errors.

III. DATASET

We use the TITAN dataset [1], an urban-driving dataset with synchronized image frames and per-object annotations sampled at 10Hz. We restrict supervision to pedestrians and to the five person-related action families used by our heads, discarding all vehicle-centric attributes. Because all experiments are run on a single workstation, we are constrained to computationally tractable settings. For person detection and tracking, we initially fine-tune on the full dataset for a single epoch, but this proved too slow to complete within the project timeline, we conduct a subsequent experiment with a 10% subset. In contrast, action recognition is trained and evaluated from the outset on a 20% subset of clips.

A. Detection Splits for YOLO11

For person detection we convert TITAN person boxes into YOLO format. The full (100%) detection split and a 10% subset are summarized in Table I. YOLO11n is fine-tuned on 100% of this split for a single epoch to probe the benefit of full dataset coverage, and on the 10% subset for a longer 30-epoch schedule.

Split	# images		# person boxes	
	100%	10%	100%	10%
Train	40,032	4,396	221,760	30,699
Val	21,064	3,134	119,395	21,875
Test	12,175	1,429	58,155	10,416
Total	73,271	8,959	399,310	62,990

TABLE I

SCALE OF YOLO11 DETECTION SPLITS FOR TITAN.

B. Action-Recognition Tubes and Label Distribution

For action recognition, we construct person-centric tubes by sliding a fixed-length temporal window over each ground-truth person track and cropping to the corresponding bounding box in each frame. Tubes are built directly from annotations so that the action recognizer can be evaluated independently of detection and tracking noise. In the final experiments we use windows of $T = 16$ frames with a base stride of $S = 4$.

To partially compensate for the extreme label imbalance in TITAN, we mark as *rare* any track that contains at least one label other than *none* or a walking-related label in any head. For these rare tracks we rerun the windowing with stride $S = 1$, while common tracks remain at stride $S = 4$.

On the 20% clip subset used in this work, this procedure yields **33,144** tubes for train, **24,740** for validation, and **12,212** for test, for a total of 70,096 tubes. Table II reports, for each head, the fraction of *none* labels in train/val/test and over all splits combined. The atomic head is almost fully annotated with non-*none* classes, while the contextual heads remain heavily dominated by *none* even after this rare-track oversampling.

Head	train	val	test	all
Atomic	0.2%	0.3%	0.0%	0.2%
Simple-ctx	15.7%	15.1%	21.4%	16.5%
Complex-ctx	98.2%	99.2%	99.0%	98.7%
Comm.	83.8%	81.2%	81.9%	82.6%
Transp.	89.0%	90.4%	91.9%	90.0%

TABLE II

FRACTION OF *none* LABELS PER HEAD AND SPLIT FOR THE 20% TUBE SUBSET WITH $T = 16$ AND RARE-TRACK OVERSAMPLING.

Figure 1 shows the training-set label distribution for all non-*none* classes across the five heads (log-scale y -axis). Even after oversampling rare tracks, the atomic and simple-context heads are dominated by common behaviours, while many rare actions occur only a few dozen times. For the complex-context, communicative, and transportive heads, positive tubes are concentrated in a handful of classes and remain two–three orders of magnitude fewer than the dominant atomic/simple-context modes, explaining why these heads can achieve high accuracy by predicting *none*, yet still attain poor mAP on rare actions.

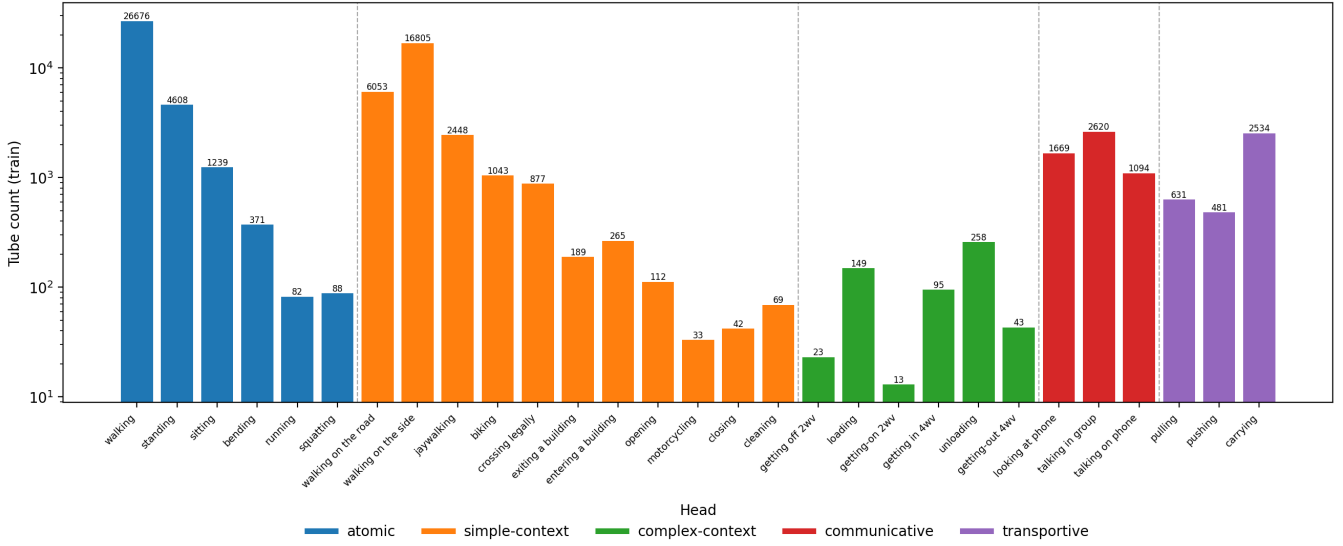


Fig. 1. Training-set label distribution over all non-*none* classes for each action head (log-scale). Colours indicate the head.

IV. METHODOLOGY

A. Detection and Tracking

1) *Detector*: We experimented with three YOLO11 configurations for single-class person detection [2], all trained or evaluated on a MacBook Pro with an Apple M4 Max and 48 GB of unified memory using the MPS backend:

- **YOLO11n-full**: YOLO11n fine-tuned on 100% of the TITAN person detection split for one epoch.
- **YOLO11n-0.1**: YOLO11n fine-tuned on the 10% subset for 30 epochs, evaluated both with and without test-time augmentation (TTA).
- **YOLO11L**: YOLO11L evaluated off-the-shelf with COCO-pretrained weights (no TITAN fine-tuning).

As discussed in Section III, we restrict the full-split experiment to a single epoch and use the 10% subset for the 30-epoch schedule. These experiments probe how well an off-the-shelf detector can localize persons on TITAN and how much benefit comes from modest fine-tuning. The detection results suggest that a properly trained YOLO11 detector on the full TITAN person split would likely deliver near-perfect pedestrian boxes.

2) *Tracker*: Our target online pipeline uses ByteTrack [3] to associate per-frame person detections into tracklets, producing MOT-style sequences of frame indices, track IDs, and bounding boxes. In the experiments reported here, we do not use tracker outputs to build tubes; instead, we construct oracle tubes directly from the ground-truth person tracks and keep the ByteTrack setup in reserve for future work that integrates predicted boxes.

B. Action Recognition from Person Tubes

Given a ground-truth person track and its per-frame bounding boxes, we first construct fixed-length tubes by cropping a temporal window $\{I_t\}_{t=1}^T$ around the box (top-left $xywh$) in each frame and resizing each crop to 112×112 . For each

tube, we stack the $T = 16$ crops into a tensor of shape (C, T, H, W) and normalize with the Kinetics-400 mean and standard deviation.

On the training set we further bias sampling and augmentation toward rare actions in three stages. First, using the empirical label distribution on the tubes constructed in Section III, we identify rare classes in each head (frequency $\leq 15\%$) and mark tubes that contain at least one such class in any head as “rare”. We then rebalance the tube list by subsampling majority tubes so that they are at most three times as numerous as rare tubes and duplicating each rare tube once. Finally, we build a weighted sampler that assigns each tube a weight inversely proportional to the frequency of its labels (aggregated across heads), with clipping to prevent a few tubes from dominating. This sampler is used with replacement to draw each epoch. During training, all rare tubes are augmented through a horizontal flip and color jitter to counter class imbalance.

1) *Backbone and Heads*: Figure 2 illustrates the architecture. Our action recognizer is a 3D ResNet-18 (R3D-18) backbone pretrained on Kinetics-400, following the spatio-temporal residual architecture of Hara *et al.* [4]. The 3D convolutional trunk and spatio-temporal pooling reduce an input tube to a global feature vector $\mathbf{x} \in \mathbb{R}^{512}$. We discard the original classification layer and attach **five** separate multilayer heads, one per TITAN action family (*atomic*, *simple-context*, *complex-context*, *communicative*, and *transportive*). Each head h applies the transformation

$$\mathbf{z}^{(h)} = W_2^{(h)} \sigma(W_1^{(h)} \mathbf{x} + \mathbf{b}_1^{(h)}) + \mathbf{b}_2^{(h)}, \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{512}$ is the shared feature vector, $W_1^{(h)} \in \mathbb{R}^{256 \times 512}$, $W_2^{(h)} \in \mathbb{R}^{C_h \times 256}$, $\sigma(\cdot)$ denotes a ReLU followed by dropout, and C_h is the number of classes (including *none*) for head h .

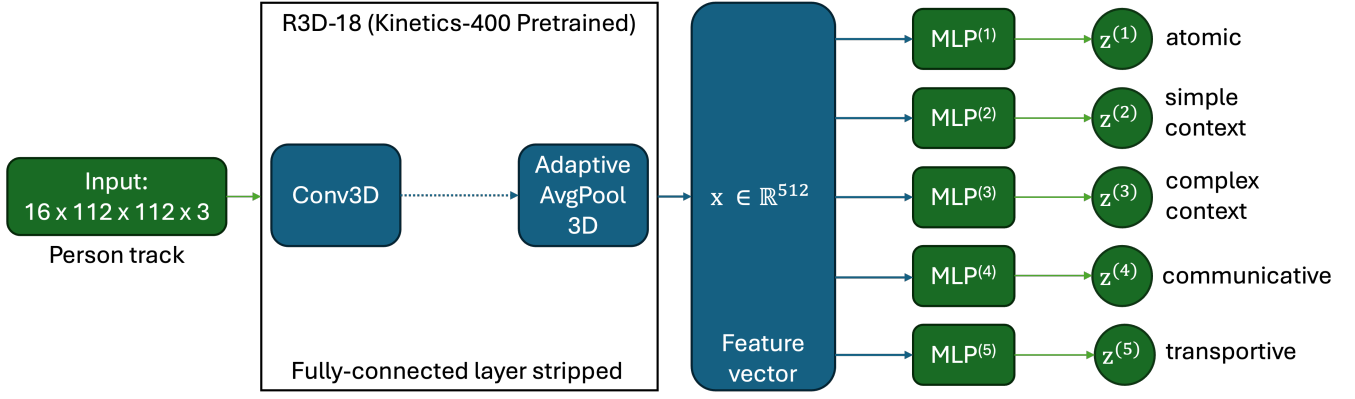


Fig. 2. Overview of the multi-head action recognition architecture. An input person tube of size $16 \times 112 \times 112 \times 3$ is passed through an R3D-18 backbone pretrained on Kinetics-400. For clarity, only the first 3D convolutional layer and the final global spatiotemporal pooling layer are shown; the dotted block represents the remaining residual layers of R3D-18. The resulting 512-dimensional feature vector $\mathbf{x}$ is fed into five independent MLP heads $\text{MLP}^{(h)}$, each producing logits $\mathbf{z}^{(h)}$ for one TITAN action family.

2) *Loss*: Let $\mathcal{H}$ denote the set of the five action heads (atomic, simple-context, complex-context, communicative, and transportive), with $|\mathcal{H}| = 5$. For each head $h \in \mathcal{H}$ and tube with feature vector $\mathbf{x}$, the corresponding MLP produces logits $\mathbf{z}^{(h)} \in \mathbb{R}^{C_h}$ and probabilities

$$\begin{aligned} p_c^{(h)} &= p(y^{(h)} = c | \mathbf{x}) \\ &= \frac{\exp(z_c^{(h)})}{\sum_{j=1}^{C_h} \exp(z_j^{(h)})}, \end{aligned} \quad (2)$$

for $c = 1, \dots, C_h$.

From the training label histogram of head h we derive inverse-frequency class weights $w_c^{(h)} > 0$, normalized so that $\frac{1}{C_h} \sum_c w_c^{(h)} \approx 1$. For the *atomic* and *simple-context* heads we use weighted cross-entropy with label smoothing $\epsilon = 0.05$. Writing $y^{(h)}$ for the ground-truth class index and

$$q_c^{(h)} = (1 - \epsilon) \mathbb{I}[c == y^{(h)}] + \frac{\epsilon}{C_h}, \quad (3)$$

the per-sample loss for head h is

$$L_h^{\text{CE}} = - \sum_{c=1}^{C_h} w_c^{(h)} q_c^{(h)} \log p_c^{(h)}. \quad (4)$$

For the three harder, highly imbalanced heads

$$\mathcal{H}_{\text{focal}} = \{\text{complex-context, communicative, transportive}\}$$

we instead apply a focal-modulated, class-weighted loss without label smoothing [10]. Letting $p_{y^{(h)}}^{(h)}$ denote the predicted probability of the true class and using the same weights $w_{y^{(h)}}^{(h)}$, the per-sample focal loss is

$$L_h^{\text{focal}} = -w_{y^{(h)}}^{(h)} (1 - p_{y^{(h)}}^{(h)})^\gamma \log p_{y^{(h)}}^{(h)}, \quad (5)$$

with focusing parameter $\gamma = 1.5$. This term down-weights easy, high-confidence examples and puts more emphasis on misclassified or rare classes.

For a mini-batch, we average L_h^{CE} or L_h^{focal} over samples to obtain the head loss L_h , and we define the overall multi-task

objective as the uniform average over heads:

$$\mathcal{L} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} L_h. \quad (6)$$

3) *Optimization*: Training is performed on a MacBook Pro with an Apple M4 Max using the MPS backend. We optimize all parameters with SGD and Nesterov momentum 0.9, weight decay 10^{-4} , batch size 24, and an initial learning rate of 3×10^{-4} . The learning rate follows a cosine-annealing schedule over a 30-epoch budget. For the first four epochs we freeze the R3D-18 backbone and train only the MLP heads, after which we unfreeze the backbone and continue end-to-end training. To improve stability under heavy class imbalance and focal loss, we clip the global gradient norm at 4.0. Early stopping is triggered when the validation loss does not improve for eight consecutive epochs; we keep the checkpoint with the lowest validation loss for test-time evaluation.

4) *Evaluation*: For each head we compute per-class average precision (AP) and head mAP (excluding the *none* class), as well as head accuracy and macro-averaged mAP/accuracy across heads. We also export precision-recall curves and confusion matrices for further analysis, and generate qualitative visualizations, including single-tube examples and multi-person frames where all annotated persons are drawn with colour-coded boxes and legends listing their ground-truth and predicted actions for non-*none* heads.

V. RESULTS

A. Detection and Tracking

1) *Quantitative Detection Results*: We benchmark the three YOLO11 configurations on the 10% TITAN person test subset from Table I (1,429 images, 10,416 person instances). Table III reports box mAP metrics.

Even a single epoch of YOLO11n on the full TITAN split yields the highest mAP, suggesting that dataset coverage matters more than longer training on a 10% subset; the larger YOLO11L nearly matches YOLO11n-full without

fine-tuning, while YOLO11n on 10% lags behind but recovers much of the gap with TTA.

Model	Data (split, ep.)	mAP ₅₀	mAP _{50:95}
YOLO11n-full	100%, 1	0.987	0.840
YOLO11n-0.1	10%, 30	0.952	0.793
YOLO11n-0.1+TTA	10%, 30	0.970	0.810
YOLO11L	100%, 0	0.970	0.838

TABLE III

PERSON DETECTION PERFORMANCE ON THE TITAN PERSON TEST SUBSET.

2) *Qualitative Comparison:* Figure 3 shows a representative frame overlaid with ground-truth boxes and detections from YOLO11n-full, YOLO11n-0.1, and YOLO11L. YOLO11n-full and YOLO11L both localize small or partially occluded pedestrians robustly, whereas the 10% YOLO11n model occasionally misses distant persons or produces slightly misaligned boxes, consistent with its lower recall.

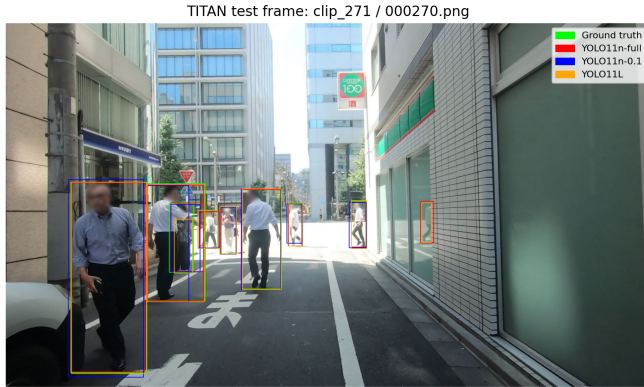


Fig. 3. Qualitative comparison of ground-truth person boxes and detections from YOLO11n-full (100%), YOLO11n-0.1 (10%), and YOLO11L on the same TITAN frame.

3) *Downstream Usage:* Given these results, we expect that a fully trained YOLO11n on the full TITAN dataset could provide high-quality person boxes for action recognition. In this report, we instead use ground-truth boxes to isolate the behaviour of the multi-head action recognizer.

B. Action Recognition

We train the R3D-18 backbone with five action heads on the 20% TITAN tube subset described in Section III, using tubes of length $T=16$ and 112×112 crops. Training follows the loss and optimization scheme of Section IV-B, with early stopping based on validation loss.

Figure 4 shows that training loss decreases steadily, while validation loss drops more slowly and then plateaus, indicating that the model continues to fit the data without severe overfitting. The inflection points around epoch 5 coincide with unfreezing the backbone; some heads briefly lose accuracy before recovering as the shared features adapt. The accuracy and mAP curves highlight that the atomic and

simple-context heads benefit most from training, whereas complex-context, communicative, and transportive heads remain dominated by the *none* class and achieve only modest gains in mAP despite high accuracy.

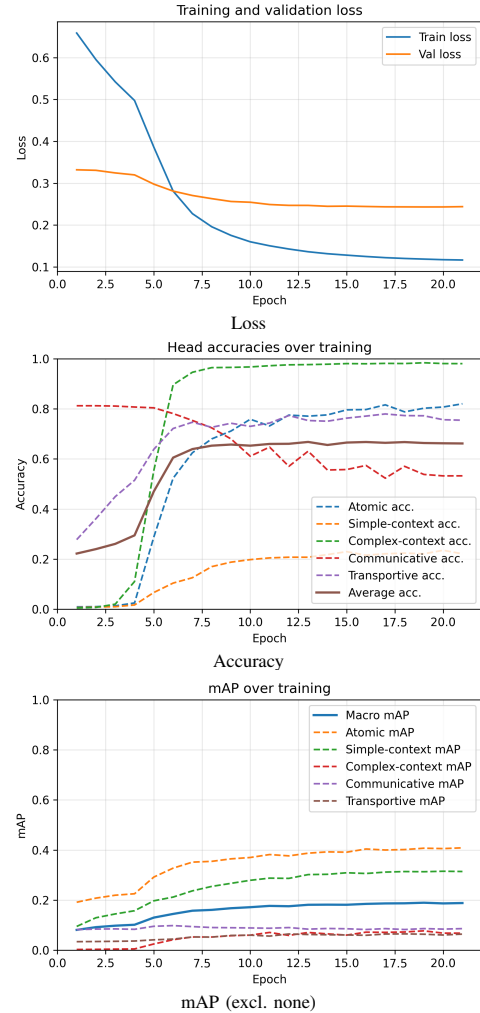


Fig. 4. Training diagnostics for the R3D-18 action recognizer: training/validation loss (top), validation accuracy per head and macro-average (middle), and validation mAP per head and macro-average (bottom).

1) *Head-wise Test Performance:* Table IV reports test accuracy and mAP for each head. As in training, mAP is computed over non-*none* classes within each head and then averaged across heads.

Head	Accuracy	mAP (excl. none)
Atomic	0.799	0.413
Simple-context	0.548	0.175
Complex-context	0.993	0.005
Communicative	0.847	0.109
Transportive	0.967	0.016
Macro average	0.831	0.144

TABLE IV

PER-HEAD TEST ACCURACY AND MAP



Fig. 5. Qualitative tube-level predictions on a TITAN test frame. Left: ground-truth person boxes with track IDs. Right: for each visible person, predicted actions for all heads with ground-truth labels in parentheses when non-*none*. The model is reliable on frequent atomic actions (e.g., *walking*) but often maps rare contextual, communicative, and transportive behaviours to *none*.

Overall, the macro-averaged test accuracy is 0.83 with a macro mAP of 0.14. The atomic head achieves the best balance of accuracy and mAP: it recognizes common motions reliably but still underperforms on rare classes. Simple-context actions are noticeably harder, with moderate accuracy and lower mAP. Complex-context and transportive heads obtain very high accuracy but almost zero mAP, indicating that the model is usually correct when predicting *none* yet rarely recovers positive loading/unloading or pulling/pushing behaviours. The communicative head lies between these extremes, with high accuracy but only modest mAP.

2) *Per-class AP*: Per-class AP values further highlight this imbalance. For the atomic head, frequent actions such as *walking*, *standing*, and *sitting* reach APs of approximately 0.95, 0.72, and 0.60, respectively, while rarer motions such as *bending* and *running* fall below 0.15 and *squatting* has essentially zero AP. In the simple-context head, *biking* (AP ≈ 0.77) and *walking on the road* (AP ≈ 0.42) are captured reasonably well, but door- and building-related interactions (*entering/exiting a building*, *opening/closing*) and *motorcycling* have APs close to zero.

3) *Qualitative Tube-level Predictions*: Figure 5 illustrates tube-level predictions on a crowded test frame. For each visible person we list the predicted label for each head and the ground-truth label in parentheses. The recognizer is generally reliable on frequent atomic actions such as *walking*, but it sometimes confuses similar motions (e.g., *walking* vs. *standing*, *sitting* vs. *bending*) and often misses contextual or communicative events, defaulting to *none*. These qualitative patterns are consistent with the per-class AP reported above.

VI. CONCLUSION AND FUTURE WORK

We presented a perception pipeline for TITAN that combines single-class pedestrian detection, tracking, and multi-head action recognition. On the detection side, YOLO11 variants achieve strong person localization on TITAN, with box mAP₅₀ up to 0.987 under modest fine-tuning, suggesting that high-quality pedestrian boxes are attainable even with limited compute. For action recognition, a multi-head R3D-18 trained on tube-based splits reaches a macro accuracy of 0.83 but only 0.14 macro mAP over non-*none* classes, despite rare-track oversampling, class-weighted losses, and focal loss on the most imbalanced heads. This gap confirms that extreme label imbalance across TITAN’s action families remains a major bottleneck for reliable recognition of rare behaviours.

Future work will focus on closing this gap and moving toward a fully deployed stack. Promising directions include training on larger subsets of TITAN or incorporating additional datasets, exploring stronger backbones or temporal models (e.g., deeper 3D CNNs or transformer-based architectures), and applying more advanced imbalance-mitigation strategies such as curriculum learning, decoupled head training, or hierarchy-aware losses. Integrating detector- and tracker-derived tubes into the training and evaluation loop, rather than relying solely on oracle tracks, is another key step toward understanding end-to-end performance in realistic urban-driving scenarios.

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